🥈 Silver Intelligence Dashboard — MCX Pro Edition

CXO Analytics Suite | Live Market Data | April 2026

Overview

A production-grade, multi-tab Streamlit dashboard for Silver (MCX) trading intelligence. Inspired by the Silver Trading Masterbook framework (Chapters 1–5, 7–8), built on **100% live market data** — no hardcoded PDF values.

Tabs

Tab	Content	Live Data Sources
📊 Overview	KPI strip, candlestick chart, signal panel, correlation matrix	Yahoo Finance (SI=F)
🔄 COMEX→MCX	Live price bridge calculator, MCX estimate chart, waterfall decomposition	Yahoo Finance (SI=F, USDINR=X)
🌐 Macro Framework	5-factor scorecard, DXY, Real Yields, yield decomposition, DXY↔Ag scatter	Yahoo Finance + FRED API
⚙️ Fundamentals	Supply/demand deficit, SLV ETF proxy, sector demand donut, mine supply risk	Silver Institute reference + SLV live
⚖️ Gold/Silver Ratio	Live GSR chart, signal ladder, relative performance chart	Yahoo Finance (SI=F, GC=F)
📈 Technical Analysis	Candlestick + BB, RSI, OBV, BB Width squeeze detector	Yahoo Finance (SI=F)

Live Data Sources

Data	Source	Ticker / Series
COMEX Silver Futures	Yahoo Finance	`SI=F`

Data	Source	Ticker / Series
COMEX Gold Futures	Yahoo Finance	`GC=F`
US Dollar Index	Yahoo Finance	`DX-Y.NYB`
USD/INR Exchange Rate	Yahoo Finance	`USDINR=X`
iShares Silver ETF (SLV)	Yahoo Finance	`SLV`
Copper Futures	Yahoo Finance	`HG=F`
Palladium Futures	Yahoo Finance	`PA=F`
10-yr TIPS Real Yield	FRED St. Louis Fed	`DFII10`
10-yr Nominal Treasury	FRED St. Louis Fed	`DGS10`
10-yr Breakeven Inflation	FRED St. Louis Fed	`T10YIE`

Note: FRED data is fetched directly (no API key required). All data is cached for 5 minutes to avoid rate limits.

Installation & Launch

```bash
# 1. Install dependencies
pip install -r requirements.txt

# 2. Launch the dashboard
streamlit run silver_dashboard.py

# 3. Open in browser (auto-opens)
# Default: http://localhost:8501
```

Key Features

- **Live COMEX→MCX Calculator** with adjustable duty/FX overrides
- **Macro Scorecard** (DXY, Real Yield, Fed proxy, USD/INR, momentum)
- **Gold/Silver Ratio signal ladder** with current level highlighted

- **Three-indicator technical suite:** RSI(14) with divergence, Bollinger Bands (20,2) with squeeze detector, OBV
- **Supply/Demand fundamentals** with Silver Institute reference data
- **Auto-refresh** every 5 minutes via Streamlit cache TTL

Architecture

```
silver_dashboard.py
├── Design tokens (dark theme, IBM Plex Mono + Syne fonts)
├── Data Layer (@st.cache_data, 5-min TTL)
│   ├── yfinance wrappers (_yf)
│   └── FRED CSV API wrappers (_fred)
├── Technical Indicators
│   ├── RSI (EWM-based, 14-period)
│   ├── Bollinger Bands (20-period, 2σ)
│   └── OBV (cumulative volume)
├── Macro Signal Engine (5-factor scoring)
├── MCX Pricing Formula
└── 6 Tabs (each independently rendered)
```

Disclaimer

Educational and analytical purposes only. Not financial advice.
Silver trading involves significant risk of loss.